Volumes by the Disc and Washer Methods

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Decompose the rational expression $\frac{5x^2-3x+2}{(x-1)(x^2+1)}$ into partial fractions.

1. Set up the partial fraction template with irreducible quadratic: $\frac{5x^2-3x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$. 2. Multiply through by $(x-1)(x^2+1)$: $5x^2-3x+2 = A(x^2+1) + (Bx+C)(x-1)$. 3. Substitute $x=1$ (cover-up): $5(1)^2-3(1)+2 = A(1+1)+0 \Rightarrow 4 = 2A \Rightarrow A = 2$. 4. Compare coefficients: Coefficient of x^2: $A+B=5 \Rightarrow 2+B=5 \Rightarrow B=3$. Constant term: $A-C=2 \Rightarrow 2-C=2 \Rightarrow C=0$. D.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 2 · MediumFind the third-degree Taylor polynomial $P_3(x)$ for $f(x)=\sqrt{x}$ centered at $c=4$.

Evaluate derivatives at $c=4$: • $f(x)=x^{1/2} \Rightarrow f(4)=2$. • $f'(x)=\frac{1}{2}x^{-1/2} \Rightarrow f'(4)=\frac{1}{2(2)}=\frac{1}{4}$. • $f''(x)=-\frac{1}{4}x^{-3/2} \Rightarrow f''(4)=-\frac{1}{4(8)}=-\frac{1}{32}$. • $f'''(x)=\frac{3}{8}x^{-5/2} \Rightarrow f'''(4)=\frac{3}{8(32)}=\frac{3}{256}$. Assemble the Taylor polynomial: $P_3(x)=2+\frac{1}{4}(x-4)+\frac{-1/32}{2!}(x-4)^2+\frac{3/256}{3!}(x-4)^3$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A student receives the following scaled marks on IB AA HL assessment: Paper 1: 85/110, Paper 2: 92/110, Paper 3: 44/55, and IA: 16/20. Calculate the student's overall weighted percentage.

1. Compute the component percentages: Paper 1: $\frac{85}{110} \approx 77.27\%$ Paper 2: $\frac{92}{110} \approx 83.64\%$ Paper 3: $\frac{44}{55} = 80.00\%$ IA: $\frac{16}{20} = 80.00\%$ 2. Multiply by component weights (30%, 30%, 20%, 20%):

$\text{Overall} = 0.30(77.27) + 0.30(83.64) + 0.20(80.00) + 0.20(80.00)$ $\text{Overall} = 23.181 + 25.091 + 16.000 + 16.000 = 80.27\% \approx 80.3\%$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 3 · Advanced / AP FRQLet $P_4(x)$ be the fourth-degree Maclaurin polynomial for $\cos(x)$. Use the Lagrange Error Bound to prove that \(|\cos(0.5) - P_4(0.5)|

For $f(x)=\cos(x)$, $f^{(5)}(x)=-\sin(x)$. Since $|-\sin(t)| \le 1$ for all t, we take $M=1$. Apply the Lagrange Error Bound formula with $n=4, c=0, x=0.5$: $|R_4(0.5)| \le \frac{M}{(4+1)!}|0.5-0|^{4+1} = \frac{1}{5!}(0.5)^5 = \frac{1}{120}\left(\frac{1}{32}\right) = \frac{1}{3840}$. Because $3840 > 3000$, it follows that: \[\frac{1}{3840}

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · Advanced / AP FRQThe alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ converges to sum S. If the sum is approximated by the fourth partial sum S_4, use the Alternating Series Error Bound to find an upper bound for the error $|S - S_4|$.

The series satisfies the hypotheses of the AST: $a_n = \frac{1}{n^4}$ is positive, strictly decreasing, and $\lim_{n\to\infty} \frac{1}{n^4} = 0$. By the Alternating Series Error Bound: $|S - S_4| \le a_5 = \frac{1}{5^4} = \frac{1}{625} = \mathbf{0.0016}$. The approximation error is guaranteed to be less than or equal to 0.0016.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left|\frac{(n+1)!\,(2x-1)^{n+1}}{n!\,(2x-1)^n}\right| = |2x-1| \lim_{n\to\infty} (n+1)$. For any $x \neq \frac{1}{2}$, this limit is ∞. The series converges only at its center $x = \frac{1}{2}$. Thus, the radius of convergence is $\mathbf{R = 0}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

Compute derivatives at $x = 0$: • $f(0) = \cos(0) = 1$ • $f'(0) = -\sin(0) = 0$ • $f''(0) = -\cos(0) = -1$ • $f'''(0) = \sin(0) = 0$ Applying the Maclaurin formula: $P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 = 1 - \frac{x^2}{2}$. (Notice the cubic term has coefficient 0).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

Use $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \ldots$ with $u = 2x$: $\ln(1+2x) = 2x - 2x^2 + \frac{8}{3}x^3 - \ldots$. Convergence requires $-1 < 2x \le 1 \Rightarrow -\frac{1}{2} < x \le \frac{1}{2}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsExpress the repeating decimal $0.272727\ldots = 0.\overline{27}$ as an infinite geometric series, and find its exact fraction representation in simplest form.

Write as a series: $0.\overline{27} = \frac{27}{100} + \frac{27}{10000} + \frac{27}{1000000} + \ldots$ First term $a = \frac{27}{100}$, common ratio $r = \frac{1}{100}$. Since \(|r| = 0.01

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

Substitute $u = \ln x$. The derivative is $du = \frac{1}{x}\,dx$. The integral simplifies directly to: $\int \cos(u)\,du = \sin(u) + C = \sin(\ln x) + C$

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Limits Graphically and Numerically

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 The Definition of the Derivative

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 The Chain Rule

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{16\pi}{3}$

Disc formula: $V = \pi\int_0^4 [R(x)]^2 dx = \pi\int_0^4 (\sqrt{x})^2 dx = \pi\int_0^4 x\,dx = \pi[\frac{x^2}{2}]_0^4 = 8\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume when the region bounded by $y = x^2$, $x = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{32\pi}{5}$

Revolving about y-axis: $x = \sqrt{y}$. $V = \pi\int_0^4 (\sqrt{y})^2 dy = \pi\int_0^4 y dy = \pi[\frac{y^2}{2}]_0^4 = 8\pi$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the cross-sectional shape of a solid formed by the disc method?

a. **A solid circular disk of area** πR^2 ✓

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

The disc method generates perpendicular slices that are solid circular disks with area $A(x) = \pi[R(x)]^2$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$ ✓

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

Intersections at $x = 0, 1$. Outer radius $R(x) = x$, inner radius $r(x) = x^2$.
$V = \pi\int_0^1 (x^2 - x^4)dx = \pi[\frac{x^3}{3} - \frac{x^5}{5}]_0^1 = \pi(\frac{1}{3} - \frac{1}{5}) = \frac{2\pi}{15}$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$ ✓

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$

Outer radius is $2x$, inner radius is x^2. Washer area is $\pi[(2x)^2 - (x^2)^2] = \pi(4x^2 - x^4)$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why is $\pi\int(R-r)^2dx$ INCORRECT for the washer method?

a. **The area of a washer is $\pi R^2-\pi r^2$, which is NOT equal to $\pi(R-r)^2$** ✓

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

Area of an annulus is $\pi(R^2-r^2)=\pi R^2-\pi r^2$. Squaring the difference $(R-r)^2=R^2-2Rr+r^2$ introduces an erroneous $-2Rr$ term.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The region bounded by $y=x^2$ and $y=4$ is revolved about the line $y=4$. What is the outer radius $R(x)$?

a. $R(x)=4-x^2$ ✓

b. $R(x)=x^2$

c. $R(x)=4$

d. $R(x)=x^2-4$

The axis of revolution is $y=4$. The distance from the curve $y=x^2$ to the axis $y=4$ is $4-x^2$, which forms solid disks with radius $4-x^2$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume in the above problem revolved about $y=4$.

a. $\frac{256\pi}{15}$ ✓

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

$V=\pi\int_{-2}^{2}(4-x^2)^2dx=2\pi\int_{0}^{2}(16-8x^2+x^4)dx=2\pi[16x-\frac{8x^3}{3}+\frac{x^5}{5}]_0^2=2\pi[32-\frac{64}{3}+\frac{32}{5}]=2\pi[\frac{480-320+96}{15}]=2\pi[\frac{256}{15}]=\frac{512\pi}{15}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The region between $y = \sqrt{x}$, $y = 0$, and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ **and** $r(x) = 1$ ✓

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$

Outer curve is $y = \sqrt{x}$, distance to $y = -1$ is $\sqrt{x} - (-1) = \sqrt{x} + 1$. Inner boundary is $y = 0$, distance to $y = -1$ is $0 - (-1) = 1$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$ ✓

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$

Side of square is $s(x) = 1 - x^2$. Area $A(x) = (1 - x^2)^2 = 1 - 2x^2 + x^4$.

$V = \int_{-1}^{1} (1 - 2x^2 + x^4)dx = 2[x - \frac{2x^3}{3} + \frac{x^5}{5}]_0^1 = 2(1 - \frac{2}{3} + \frac{1}{5}) = 2(\frac{8}{15}) = \frac{16}{15}$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$ ✓

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

Radius is $r = \frac{s(x)}{2}$. Semicircle area is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(\frac{s}{2})^2 = \frac{\pi}{8}s^2$.

© ClipSAT — clipsat.org | All Rights Reserved

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$ ✓

b. $A(x) = \frac{1}{2}[s(x)]^2$

c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

The area of an equilateral triangle with side s is $\frac{\sqrt{3}}{4}s^2$.

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. **They give the same volume** ✓

c. Washer is always larger

d. Depends on axis and region

Both methods give the same volume for the same solid.

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi\int(R-r)^2dx$

b. $\pi\int(R^2-r^2)dx$

c. $2\pi\int(R-r)dx$

d. $\pi\int R^2dx - \pi\int r^2dx$ **(same)** ✓

Applying Applications of Integration principles, Washer: $\pi\int(R^2-r^2)dx$. Therefore, the verified correct option is $\pi\int R^2dx - \pi\int r^2dx$ (same).

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2 h$

b. $\frac{1}{2}\pi r^2 h$

c. $\pi r^2 h$

d. $\frac{1}{3}\pi r^2 h$ ✓

Applying Applications of Integration principles, $V = \frac{1}{3}\pi r^2 h$. Therefore, the verified correct option is $\frac{1}{3}\pi r^2 h$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8-1)/3 = 28\pi/3$ ✓

Applying Multiple Integrals principles, $V = (4\pi/3)(2^3 - 1^3) = 28\pi/3$. Therefore, the verified correct option is $4\pi(8-1)/3 = 28\pi/3$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$ ✓

d. $\frac{16\pi}{3}$

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{8} \cdot \frac{4}{3}\pi(4)^3 = \frac{32\pi}{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$ ✓

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\,\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{2}\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 The Chain Rule

The Jacobian of the transformation $x = r\cosh u, y = r\sinh u$ is r times:

a. $\cosh^2 u - \sinh^2 u = 1$ ✓

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

Applying Multiple Integrals principles, $J = r(\cosh^2 u - \sinh^2 u) = r$. Therefore, the verified correct option is $\cosh^2 u - \sinh^2 u = 1$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Centre of mass of lamina $D = [0,1] \times [0,1]$ with density $\delta = xy$:

a. $(1/2, 1/2)$

b. $(2/3, 2/3)$ ✓

c. $(3/4, 3/4)$

d. $(1, 1)$

$m = \int_0^1\int_0^1 xy\,dy\,dx = 1/4$; $M_y = \int_0^1\int_0^1 x^2 y\,dy\,dx = 1/6$; $\bar{x} = (1/6)/(1/4) = 2/3$; similarly $\bar{y} = 2/3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Use polar coordinates to evaluate $\iint_R (x^2 + y^2)^3\,dA$, where R is the disk $x^2 + y^2 \le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$ ✓

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{6561\pi}{4}$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The double integral over region D of $f = 1$ gives:

a. **area of D** ✓

b. perimeter of D

c. volume

d. 0

Applying Multiple Integrals principles, $\iint_D 1\,dA = \text{Area}(D)$. Therefore, the verified correct option is area of D.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Triple integral $\iiint_V 1\,dV$ gives:

a. **volume of V** ✓

b. 0

c. surface area

d. 1

Applying Multiple Integrals principles, Triple integral of 1 = volume. Therefore, the verified correct option is volume of V.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use polar coordinates to evaluate $\iint_R \sqrt{x^2 + y^2}\,dA$, where R is the annulus $1 \le x^2 + y^2 \le 4$.

a. $\frac{14}{3}$

b. $\frac{14\pi}{3}$ ✓

c. $\frac{28\pi}{3}$

d. $\frac{7\pi}{3}$

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{14\pi}{3}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Optimize area of rectangle with perimeter 20: max area $=$

a. 25 ✓

b. 20

c. 50

d. 100

Applying Application of Derivatives principles, 5×5 square: area $= 25$. Therefore, the verified correct option is 25.

© ClipSAT — clipsat.org | All Rights Reserved